

Nicholas De Raj

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PROFESSIONAL EXPERIENCE

Full Stack Developer, [Goldman Sachs](#)

London (May 24 - Current)

- Led a team of 4 engineers to architect and deliver a real-time **ETL** pipeline migrating data flows from Postgres to **Kafka** and **Elasticsearch**, uplifting multiple legacy services to support low-latency, scalable data publishing.
- Integrated AI-driven document processing capabilities into client onboarding workflows, automating field extraction and validation against corporate documentation to significantly reduce manual review time
- Built **Java microservices**, connected asynchronously via **Kafka** pipelines and synchronously via API calls. Mostly using **Mongo** with **CSFLE encryption** to store customer information, and with some support scripts written in **Python**.
- Used AI Tools to build **automated agent** skills that fix code quality across all 40 repos in our stack continuously in parallel.

Quant Developer, [Tokka Labs](#)

Singapore (Jan 24 - Apr 24)

- Created a **market maker system** that would publish its **orderbooks** to aggregated market makers like Uniswap, mostly for smaller cap crypto tokens but also Bitcoin, Ethereum and Solana.
- Designed a **microservice** based monorepo with **TypeScript** on **NestJS** which went live from scratch in 2 months, making our first trade on mainnet and supported over 30 tokens and millions in volume within a month after that.

Full Stack Developer, [Cake DeFi](#)

Singapore (Sep 23 - Dec 23)

- Led the implementation of an **ERC-1167** Smart Contract in **Solidity** to our multi-signatory wallet business, deploying them across multiple chains including *Ethereum*, *Polygon*, *AVAX*, and others to reduce costs by roughly 60%.

Tech Lead, [Ethlas](#)

Singapore (Sep 22 - Sep 23)

- **Led a team** of engineers, product managers and designers, to build and scale a mobile game to 100k users in under 3 months using **React Native + TypeScript**, **NodeJS** and **Google Cloud**, deployed to both IOS and Play Store.

Full Stack Developer, [Supa](#)

Kuala Lumpur (Aug 21 - Sep 22)

- Built fully functional **NextJS** apps in **TypeScript** from scratch to production, that connect to serverless **NodeJS**, **Ruby on Rails** and **Golang** microservices hosted on AWS to enable data labelling for ML systems to ingest.

Deputy CEO, [Gryphon Energy](#)

Kuala Lumpur (Apr 17 - Sep 20)

- Ran the German Power Division (Gaea Power GmbH, co-founded then injected into the group): **P&L** and O&M of eight palm-oil **CHP** units supplying heat to greenhouses and power to the grid.
- Locked in physical **palm oil** supply on **forward** contracts, took delivery into the plants, and managed fuel cost against heat and power offtake.
- Supported the oil and gas division's asset-leasing book.

PROJECTS

Quant Research & Risk Engine | [GitHub](#) | [Live Demo](#) | **Python**, GARCH/EVT, backtesting, statistics

- Designed and ran a systematic research programme across crypto, commodities, and equity futures, using pre-registered, **walk-forward** validation to rule out false positives.
- Isolated one statistically robust result, **GARCH-t/EVT tail-risk calibration**, and validated it across asset classes and holdout periods. Productionized the result as a live risk engine computing **VaR** and **Expected Shortfall**.

Commodity Futures Spread Trading System | **Python**, quantitative research, risk management, automated infrastructure

- Following on from the research above, design a 5-spread **commodity futures mean-reversion** book: **+49.0% / Sharpe 0.68** (2010–2026, engine that set the universe); held out **2024–2026: +14.9% / Sharpe 1.14**. Tune-sample hit rate **~70%** on 470+ trades.
- Traced a **-11.3% paper drawdown** to one unstopped trade and rebuilt the risk layer (**ATR** stop sizing, adverse-excursion stops, per-spread calibration).
- Automated the data-to-execution pipeline, running currently unattended in production (**paper-trading**).

SKILLS

- **Languages:** Java, Python, TypeScript/JavaScript, Solidity, SQL, Bash
- **AI Tools:** Claude Code, Codex, Cursor, Hermes Agents
- **Frontend:** React, NextJS, TypeScript
- **Backend & Systems:** NodeJS, NestJS, microservices, REST/GraphQL, Kafka, Elasticsearch, Postgres, Redis, Docker, AWS/GCP
- **Quant Research & Trading:** GARCH/EVT, time-series analysis, backtesting, walk-forward validation, VaR/Expected Shortfall, mean-reversion, spread trading, ATR stops, market microstructure, crypto market making
- **Data Engineering:** ETL pipelines, streaming (Kafka)
- **Tools:** Git/GitHub, CI/CD, Claude Code, Jupyter, Linux/Unix, monitoring & alerting
- **Leadership:** Technical lead (leading up to teams of 7), cross-functional teams, production system delivery, code quality at scale (40+ repos)

EDUCATION

UNIVERSITY COLLEGE LONDON (UCL)
Master of Science in **Financial Mathematics**

(2026 - Current Part Time)

UNIVERSITY OF BRISTOL
Bachelor of Science in **Economics and Maths** (2:1 with Hons)

(2013 - 2016)